

# Customer Churn Analysis Report

## Executive Summary

### 1. Objective

The objective of this project is to analyze customer churn behavior in a telecom company and identify the key drivers behind customer attrition. The goal is to provide actionable insights that help improve customer retention and reduce revenue loss.

### 2. Business Problem

Customer churn is one of the most critical challenges in subscription-based businesses. This analysis aims to answer the following questions:

- Which customers are most likely to churn?
- What are the main factors driving churn?
- Which customer segments are at the highest risk?

### 3. Methodology

This project was conducted using a multi-tool analytics approach:

- **Python (Pandas, Matplotlib)** → Data cleaning, EDA, feature engineering
- **SQL (PostgreSQL)** → KPI analysis and aggregation
- **Power BI** → Interactive dashboards and customer segmentation

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## Key Findings

### 1. Contract Type is the strongest churn driver

Customers with **month-to-month contracts exhibit significantly higher churn propensity**, indicating that lower contractual commitment is strongly associated with increased customer

attrition. In contrast, long-term contracts demonstrate substantially improved retention, highlighting the importance of contract duration in customer loyalty.

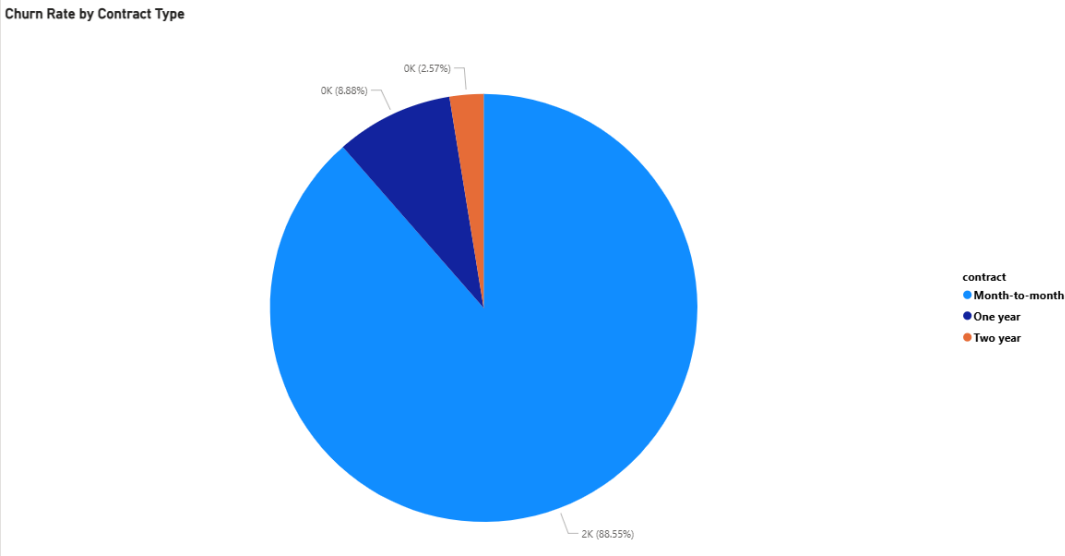


Figure 1. Churn Rate by Contract Type

2. Tenure has a strong impact

Customers within their **first 0–12 months of tenure exhibit the highest churn likelihood**, indicating a critical early lifecycle period where retention risk is significantly elevated.

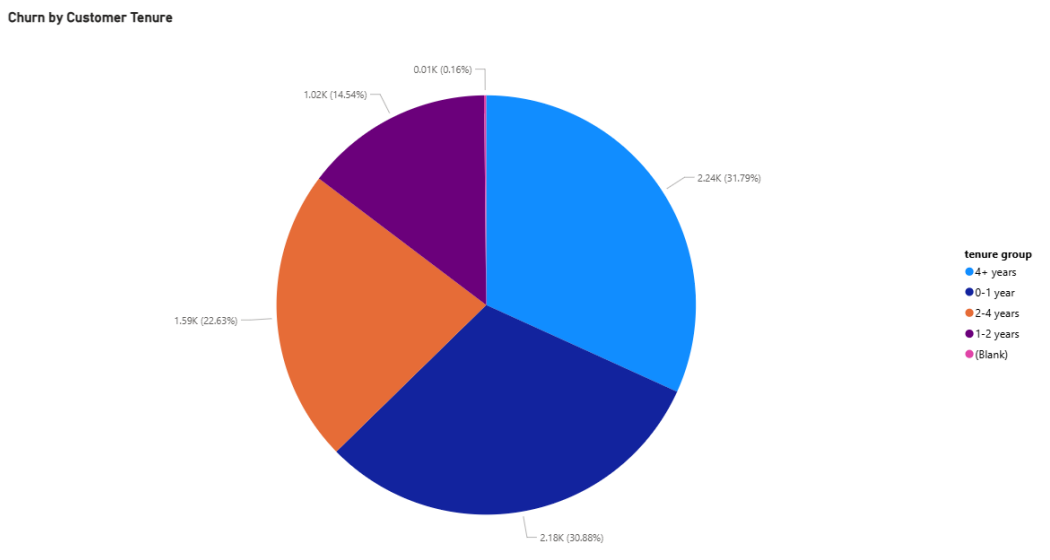


Figure 2. Churn Rate by Customer Tenure

### 3. Payment Method matters

Customers using **electronic check payments** have a much higher churn probability compared to automated payment methods.

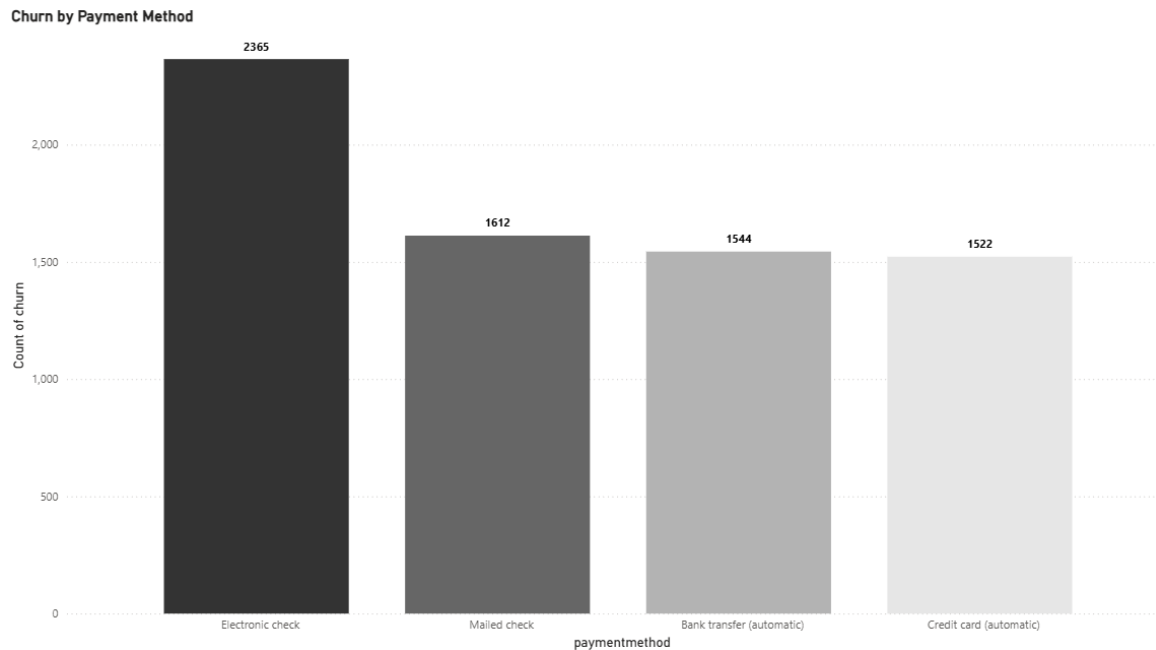


Figure 3. Churn by Payment Method

### 4. Churn by Internet Service Type

The analysis shows clear differences in churn across internet service categories.

- **Fiber Optic customers** have the highest churn count, indicating a high-risk segment despite being a premium service.
- **DSL customers** show moderate churn levels, suggesting relatively better retention.
- Customers with **No Internet Service** have the lowest churn, representing the most stable group.

Internet service type is a significant driver of churn, with Fiber Optic customers requiring the highest retention focus.

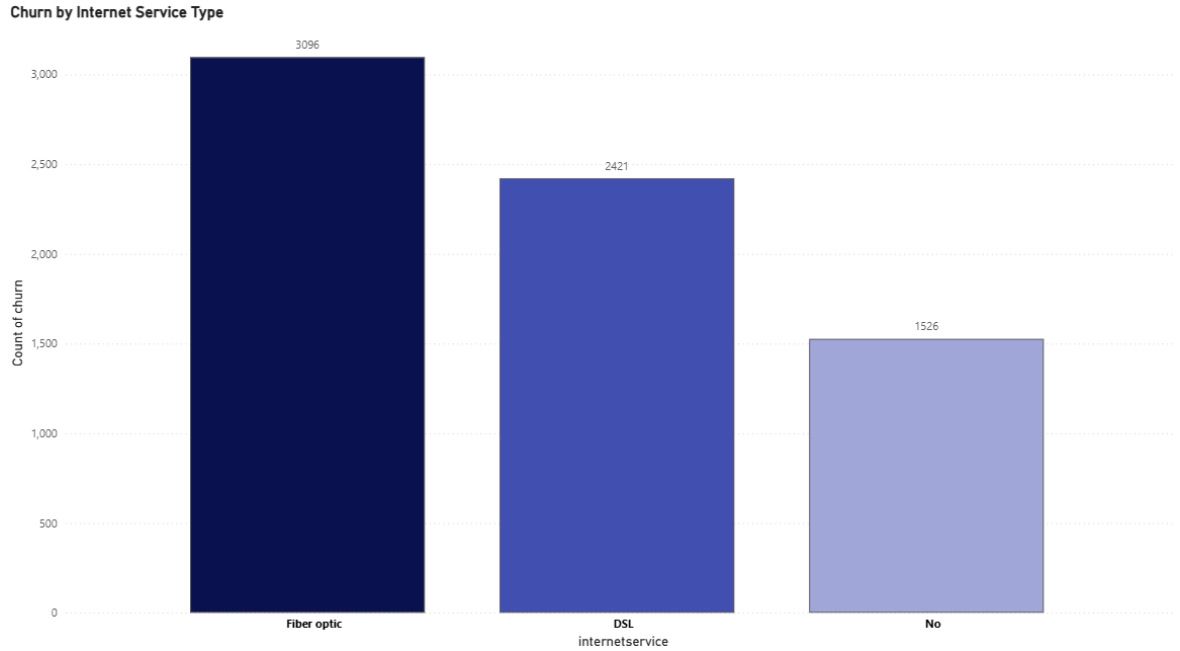


Figure 4. Churn by Internet Service Type

## 5. Value-added services reduce churn

Customers who do not subscribe to **Tech Support** and **Online Security** services exhibit a **significantly higher churn propensity**. This indicates that the absence of value-added and protective services reduces customer engagement and perceived service value, making customers more vulnerable to switching behavior. Strengthening adoption of these services can therefore play a key role in improving retention and increasing customer loyalty.

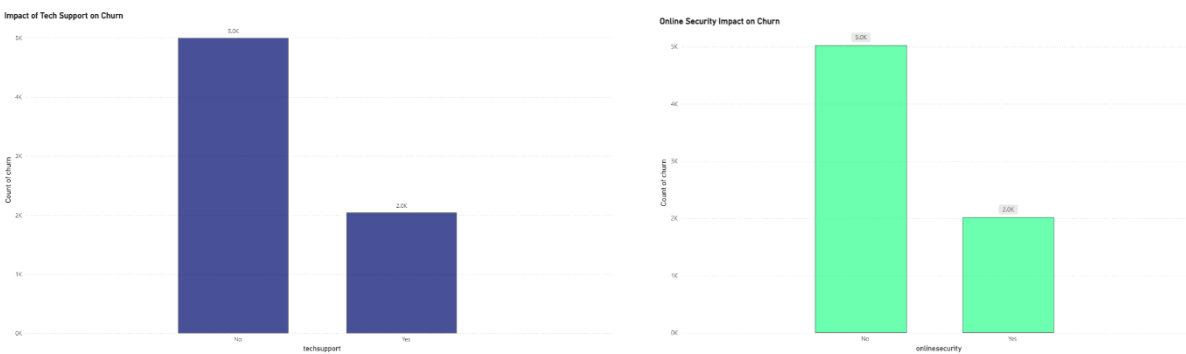


Figure 5. Impact of Tech Support and Online Security on Churn

## 5. High-value customers are at risk

A significant portion of revenue is exposed in the **High Value + High Risk segment**, making it a priority for retention strategies.

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### Customer Segmentation Insight

The segmentation analysis revealed:

- High Risk + High Value customers represent the **most critical revenue loss segment**
- Low Risk customers are highly stable and generate consistent revenue
- Risk level is a stronger predictor of churn than customer value alone

| risk_level  | High Value | Low Value | Mid Value | Total |
|-------------|------------|-----------|-----------|-------|
| High Risk   | 1.00       | 1.00      | 1.00      | 1.00  |
| Medium Risk | 1.00       | 1.00      | 1.00      | 1.00  |
| Low Risk    | 0.00       | 0.00      | 0.00      | 0.00  |
| Total       | 0.16       | 0.37      | 0.23      | 0.27  |

*Figure 6. Customer Churn Segmentation*

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### Business Recommendations

- **Improve Early Customer Experience**

Focus on onboarding and engagement during the first 12 months.

- **Promote Long-Term Contracts**

Encourage customers to switch from month-to-month to annual or two-year contracts.

- **Optimize Payment Methods**

Incentivize automated payment methods such as credit card or bank transfer.

- **Bundle Services**

Combine Tech Support and Online Security into service packages to increase retention.

- **Prioritize High-Value Customers**

Implement targeted retention strategies for high-value, high-risk customers.

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### **Expected Business Impact**

If implemented, these strategies are expected to:

- Reduce churn rate significantly
- Increase customer lifetime value (CLV)
- Improve revenue stability
- Strengthen customer loyalty in early lifecycle stages